1. On the Plausibility of Conditional Exchangeability of 401(k) Eligibility

This question asks whether 401(k) eligibility can be treated as conditionally exchangeable given income. That is, can we assume that, once income is controlled for, eligible and ineligible individuals are comparable as if eligibility were randomly assigned?

In the early days of 401(k) plans, eligibility was determined by employers and not typically a factor in job choice. Most people chose jobs based on salary and benefits more broadly. This makes it reasonable to assume that, after adjusting for income, eligibility is not strongly influenced by unobserved traits like individual saving preferences.

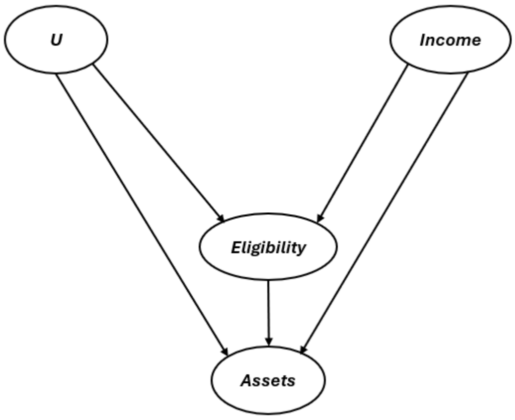

Fig. 1. Causal diagram illustrating relationships between eligibility, income, assets, and unmeasured confounding.

A simplified causal diagram (Fig. 1) helps clarify this: income affects both eligibility and net financial assets, and may proxy for deeper traits like savings preference. If conditioning on income (and possibly other covariates) blocks these backdoor paths, then eligibility becomes independent of potential outcomes, which satisfies the exchangeability assumption. This assumption underpins the validity of using methods like AIPW and TMLE to estimate the effect of eligibility on financial assets.

2. Estimating the Average Treatment Effect of 401(k) Eligibility

To estimate the effect of 401(k) eligibility (e401) on net financial assets (net_tfa), we used two doubly robust methods: Augmented Inverse Probability Weighting (AIPW) and Targeted Maximum Likelihood Estimation (TMLE). Both combine outcome regression and propensity score modeling to reduce bias and improve efficiency.

We applied the SuperLearner algorithm to estimate the nuisance functions. The ensemble included SL.glm, SL.glmnet, and SL.ranger. Covariate adjustment was based on pre-treatment variables: age, income, family size, education, gender, marital status, dual-earner status, DB pension coverage, IRA participation, and home ownership. The post-treatment variable p401 (actual 401(k) participation) was excluded to avoid conditioning on a mediator.

The AIPW estimate of the average treatment effect was $7,889.54, with a 95% confidence interval of [$5,968.22, $9,810.87]. The TMLE estimate was identical due to model convergence. For comparison, the naive difference in means without adjustment was $19,559.34, substantially higher.

This suggests that unadjusted comparisons overstate the effect due to confounding. The adjusted estimate is smaller but still statistically and economically significant, indicating that eligibility for a 401(k) plan has a meaningful positive effect on household financial assets. Although we did not implement explicit cross-fitting, the use of SuperLearner within the TMLE framework helps guard against overfitting and supports the credibility of the result.

3. Testing for Conditional Association Between Eligibility and Financial Assets

We tested whether 401(k) eligibility (e401) is conditionally associated with net financial assets, after adjusting for observed covariates. Specifically, we estimated $\theta = E\{Cov(A, Y \mid L)\}$ using the efficient influence function approach, with 5-fold cross-fitting and SuperLearner to estimate the nuisance components $E[A|L]$ and $E[Y|L]$.

The estimated value of θ was $1,673.32, with a standard error of $262.81, resulting in a z-score of 6.37 and a p-value of 1.93×10^{-10}. This provides strong statistical evidence that eligibility for a 401(k) is not conditionally independent of financial assets, even after controlling for factors like income, education, and household characteristics.

This approach is particularly valuable because it does not rely on strict functional form assumptions and remains valid under flexible machine learning estimation. The influence function also allows us to quantify uncertainty without requiring bootstrap or asymptotic approximations. Together, these elements support the credibility of our result and reinforce the role of eligibility as a potentially important driver of household saving behaviour.

4. Estimating Heterogeneous Effects of 401(k) Eligibility on Net Financial Assets

I used an R-learner approach to estimate how the effect of 401(k) eligibility on net financial assets varies across individuals. The analysis controlled for a rich set of covariates, with heterogeneity explored along age, income, and education. A five-fold cross-fitting procedure was used to estimate the nuisance components (propensity scores and outcome regressions) in a way that minimizes bias. Individual treatment effects (CATEs) were then estimated using residualized outcomes and treatment indicators, followed by weighted regression on covariates. A conservative SuperLearner library (SL.mean, SL.glm) was chosen to ensure model stability and prevent convergence issues.

All observations received valid CATE predictions (no missing estimates). The estimated propensity scores ranged from 0.10 to 0.97, indicating good overlap between treated and untreated groups. This ensures stable residuals and valid identification of treatment effects.

The histogram (Fig. 1*a*) of estimated CATEs showed a right-skewed distribution, suggesting variation in individual gains from eligibility. The average estimated treatment effect was $5,653, with a 95% confidence interval of [$5,451, $5,856]. This implies a positive and statistically significant gain in financial assets associated with 401(k) eligibility.

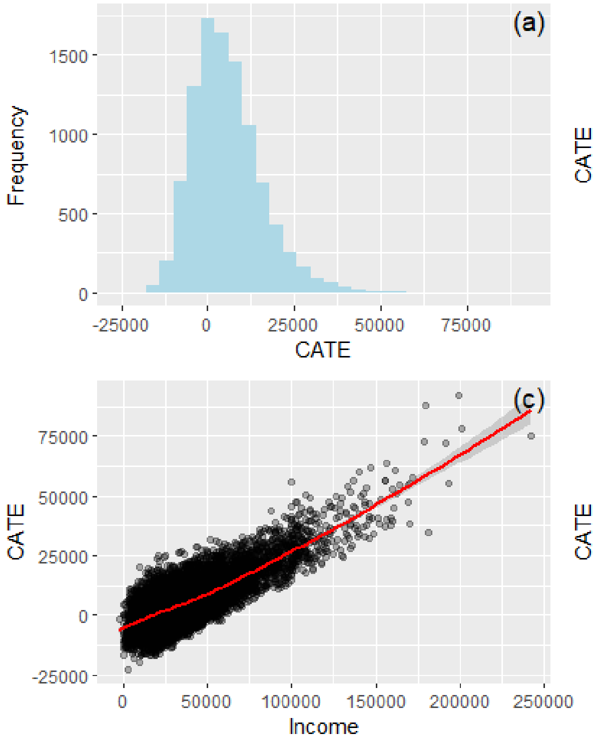

Fig. 2. Combined visualization of estimated treatment effects of 401(k) eligibility. Panel *(a)* shows the distribution of individual treatment effects (CATEs). Panels (*b), (c),* and *(d)* display marginal associations of CATEs with age, income, and education, respectively.

Heterogeneity by Key Covariates

Treatment effects varied across individual characteristics, they increased with age, suggesting greater benefits for older individuals closer to retirement (Fig. 2b). Effects also rose sharply with income (Fig. 2c), indicating higher responsiveness among wealthier individuals. For education, the pattern was nonlinear (Fig. 2c), effects dipped at moderate education levels but increased again among those with higher attainment, possibly reflecting differences in financial awareness or access to savings programs (Fig. 2).

The results suggest that 401(k) eligibility has a meaningful and statistically significant effect on net financial assets, with an average gain of approximately $5,653 per individual. Moreover, the impact is not uniform it varies systematically by age, income, and education. These findings highlight the importance of considering heterogeneity when evaluating retirement policies and suggest that eligibility incentives may be especially effective among older, higher-income, and more educated individuals.

5. Using DR-Learner for Predicting Y^1

To estimate counterfactual net financial assets under 401(k) eligibility (Y^1), a DR-learner using 5-fold cross-fitting was applied. The outcome model was trained on treated individuals using only age, income, and education, as required. Propensity scores were estimated using the full set of covariates. We then constructed pseudo-outcomes using the doubly robust formula and predicted Y^1 across the sample.

The distribution of predicted Y^1 values is shown in Figure 3a. Most values fall within a plausible range, with less than 0.1% of predictions reaching the ±$1M cap, suggesting good numerical stability. We evaluated prediction error by comparing pseudo-Y^1 to $\hat{Y}^1$, yielding a mean squared error (MSE) of $0.72 million, with a 95% bootstrap confidence interval of [$0.68M, $0.77M]. The median absolute deviation was approximately $3,300.

Diagnostics (Fig. 3b) show that propensity scores were well distributed and trimmed between 0.1 and 0.9, indicating sufficient overlap between treated and untreated individuals. Furthermore, inverse probability weights remained stable, with no values exceeding 10 and a median of 0, as expected when estimating Y^1 for untreated cases (Fig. 3c).

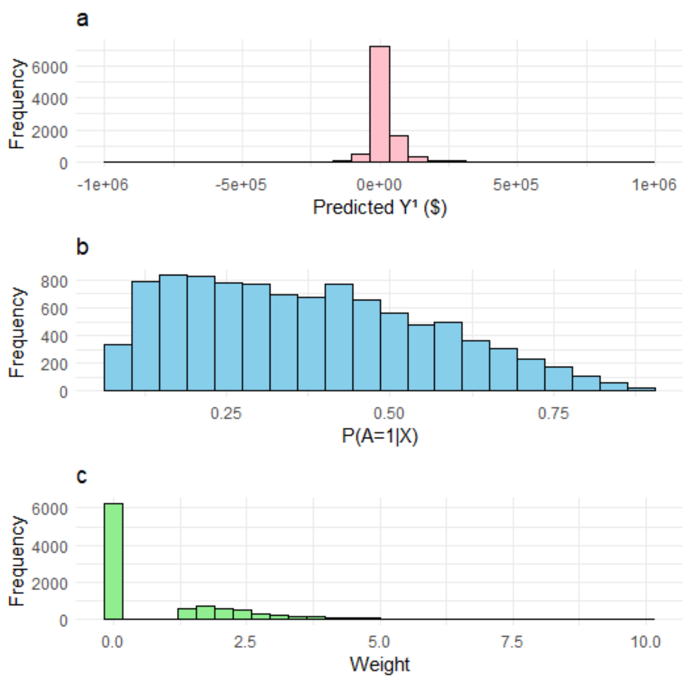

Fig. 3. Diagnostic plots from the DR-learner analysis. Panel (a) shows the distribution of predicted counterfactual outcomes ($\hat{Y}^1$) under 401(k) eligibility. Panel (b) displays the distribution of estimated propensity scores, trimmed between 0.1 and 0.9 to ensure overlap. Panel (c) shows the distribution of inverse probability weights (A/ps), all of which remain below 10, indicating good weight stability.

However, the distribution of predicted Y^1 (Fig. 3a) appeared unexpectedly concentrated near zero. This might indicate underfitting in the outcome model, potentially due to the limited covariates or the simplicity of the SuperLearner library (SL.mean, SL.glm) used for stability. While diagnostics for weights and overlap were acceptable, the central concentration suggests that future implementations should consider richer or better-tuned learners to improve prediction realism and better reflect economic behaviour.

The DR-learner framework yielded a stable estimate of counterfactual outcomes under 401(k) eligibility, with well-behaved weights and overlap. However, the predicted outcomes appeared overly concentrated near zero, likely due to underfitting from limited covariates and conservative learners. Future work should incorporate more flexible algorithms such as SL.xgboost or SL.earth to better capture individual-level variation and produce more realistic economic predictions.